

1. INTRODUCTION

The mathematical analysis of suspensions of small particles sedimenting in a Stokes flow has been the subject of extensive research over the last years. In the context of inertialess particles, a classical model is the following. Given a finite or infinite collection of identical non-overlapping spherical particles $(B_i = B(X_i, R))_{i \in I}$, set in a bounded or unbounded domain Ω , one considers the system

$$(1) \quad \begin{aligned} -\mu \Delta u + \nabla p &= f && \text{in } \Omega \setminus (\bigcup_i B_i), \\ \operatorname{div} u &= 0 && \text{in } \Omega \setminus (\bigcup_i B_i), \\ u|_{\partial\Omega} &= 0, \\ D(u) &= 0 && \text{in } B_i \quad \forall i, \\ \int_{\partial B_i} \Sigma_\mu(u, p) n d\sigma &= -mge, \quad \int_{\partial B_i} [\Sigma_\mu(u, p) n] \times (x - X_i) d\sigma(x) = 0, && \forall i. \end{aligned}$$

The unknowns of this system are the velocity field and pressure:

$$u(x) = (u_1, u_2, u_3)(x_1, x_2, x_3), \quad p(x) = p(x_1, x_2, x_3).$$

The first three lines describe a Stokes flow with viscosity μ outside the particles, with the usual no-slip condition at $\partial\Omega$. The fourth one expresses rigidity of the velocity field of each particle B_i , and is equivalent to

$$u = v_i + \omega_i \times (x - X_i) \quad \text{in } B_i$$

for some unknown translational and angular velocities $v_i, \omega_i \in \mathbb{R}^3$. The two relations in the last line, correspond to balances of forces and torques, and involve the Newtonian stress tensor $\Sigma_\mu(u, p) = 2\mu D(u) - pI_3$ with $D(u) = \frac{1}{2}(\nabla u + \nabla^\top u)$ the symmetric gradient of u . The constants v_i, ω_i can be seen as Lagrange multipliers associated with these constraints. The term $-mge$, $e = (0, 0, 1)^t$ encodes gravity and buoyancy effects: denoting ρ_p and ρ_f the particle and fluid masses per unit volume, one has $m = \frac{4}{3}\pi R^3(\rho_p - \rho_f)$, while g is the usual gravitational constant. This is the term responsible for sedimentation.

Roughly, all mathematical studies of this model divide into two categories:

- Those that focus on steady properties of the suspension. Time evolution of $X_i = X_i(t)$ is neglected (or frozen), the particle distribution is considered as given, and mean properties of the suspension, such as the average settling velocity [20, 19, 4, 9] or the effective viscosity [18, 30, 12, 13, 14, 15, 8], are deduced. The derivation of the mean settling velocity, and its expansion in terms of the solid volume fraction are notoriously difficult, and rely on subtle cancellation mechanisms which require strong assumptions on the distribution of the particles (either periodic-like or random stationary with strong mixing properties).

- Those that analyze the dynamics of the suspension (for a large but finite $N \gg 1$ collection of balls $B_1, \dots, B_N$): the time dependence of $X_i = X_i(t)$ is taken into account through coupling of (1) with the ODE

$$(2) \quad \dot{X}_i = v_i.$$

as reference length, velocity and pressure, we can put system (1) in dimensionless form:

$$\begin{aligned} X' &= X/L, \quad \varepsilon' = \varepsilon/L, \quad R' = R/L, \\ u' &= u/U, \quad p' = p/P, \quad f' = \frac{L^2}{\mu U} f, \quad X'_i = X_i/L, \quad B'_i = B_i/L = B(X'_i, R'). \end{aligned}$$

After dropping the primes, we end up with

$$\begin{aligned} (4) \quad & -\Delta u + \nabla p = f \quad \text{in } \Omega^\varepsilon \setminus (\bigcup B_i), \\ & \operatorname{div} u = 0 \quad \text{in } \Omega^\varepsilon \setminus (\bigcup B_i), \\ & u|_{\partial\Omega^\varepsilon} = 0, \\ & D(u) = 0 \quad \text{in } B_i, \quad 1 \leq i \leq N, \\ & \int_{\partial B_i} \Sigma(u, p) n d\sigma = -\frac{1}{N} e, \quad \int_{\partial B_i} \Sigma(u, p) n \times (x - X_i) d\sigma(x) = 0, \quad 1 \leq i \leq N, \end{aligned}$$

where $\Sigma(u, p) = 2D(u) - pI$. Note that we could have considered some variation of (4) with additional forcing $\int_{B_i} f$ and torque $\int_{B_i} f \times (x - X_i)$ included in the last two relations. We want to understand the asymptotic behaviour of the solution $u = u^{N, \varepsilon}$ for large N and small ε . We make three main assumptions :

- We denote by ρ^N the empirical measure defined by $\rho^N = \frac{1}{N} \sum_i \delta_{X_i}$ and consider a bounded density ρ compactly supported in $\overline{\Omega^0}$.
- We assume that there exists $\theta > 1$ such that

$$(5) \quad \varepsilon > \theta R,$$

which is a slight reinforcement of the non-overlapping constraint between the spheres and the wall.

- Denoting $d_{\min} = \min_{i \neq j} |X_i - X_j|$, we assume that

$$(6) \quad d_{\min} \geq \max\left(\frac{C}{N^{1/3}}, 2\theta R\right).$$

REMARK 1.1. — The three assumptions above, notably the lower bound (5) on the size of the depletion layer, are enough to prove our Theorems 1.1 and 1.2 and Corollary 1.3. But, as will be discussed in Section 5.1, in order to relate these mathematical results to the apparent slip phenomenon, one needs a more stringent condition than (5), essentially

$$\varepsilon \geq \mathcal{C}(R + \rho + \|\rho^N - \rho\|),$$

where $\mathcal{C}$ is a large enough constant independent of N and R , and where $\|\cdot\|$ is a weak norm, see (37). This stronger hypothesis will be discussed in detail in Section 5 as well as the phenomenon of intrinsic convection.

Under these assumptions, we will perform a two-step analysis of (4).

Step 1 : approximation by a continuous model. — We will first show that the solution $u^{N,\varepsilon}$ of (4) can be approximated by the continuous solution u^ε of

$$(7) \quad \begin{aligned} -\Delta u^\varepsilon + \nabla p^\varepsilon &= f - \rho e & \text{in } \Omega^\varepsilon, \\ \operatorname{div} u^\varepsilon &= 0 & \text{in } \Omega^\varepsilon, \\ u^\varepsilon|_{\partial\Omega^\varepsilon} &= 0. \end{aligned}$$

We assume in all what follows that $f \in L^1(\Omega^\varepsilon) \cap L^\infty(\Omega^\varepsilon)$. In particular, $f \in L^{6/5}(\Omega^\varepsilon)$ so that (4), resp. (7), has a unique solution $u^{N,\varepsilon}$, resp. u^ε in

$$V = \{u \in H_{\text{loc}}^1(\overline{\Omega^\varepsilon}), \nabla u \in L^2(\Omega^\varepsilon)\}.$$

We will prove in Section 3:

THEOREM 1.1. — *Let $1 < q < 3/2$, $Q \Subset \overline{\Omega^\varepsilon}$. There exists $C > 0$ depending on q , Q and on the constant θ in (5), such that ,*

$$(8) \quad \|u^{N,\varepsilon} - u^\varepsilon\|_{L^q(Q)} \leq C(\phi + \|\rho^N - \rho\|_{(W^{2,q'}(\Omega^0))^*} + R),$$

where $\phi = R^3 N$ the particles volume fraction.

– The norm $\|\rho^N - \rho\|_{(W^{2,q'}(\Omega^0))^*}$ can be replaced with $W_1(\rho^N, \rho)$, the first Wasserstein distance between ρ^N and ρ , and consequently by $W_p(\rho^N, \rho)$ for any $p \in [1, +\infty]$ if one assumes $\rho \in \mathcal{P}(\Omega^0)$. See Remark 3.2.

– The control on the minimal distance (6) can be relaxed to $d_{\min} > 2\theta R$ together with any assumption ensuring a uniform bound on

$$\frac{1}{N} \sum_{j \neq i} \frac{1}{|X_i - X_j|^2} \leq C,$$

see Remark 3.1 for more details.

In cases without boundaries, $\Omega = \mathbb{R}^3$ or $\Omega = \mathbb{T}^3$, this kind of statement is already known, see for instance [24]. The proof is based there on an approximation of $u^{N,\varepsilon}$ made of so-called point Stokeslets and point Stresslets, centered at the X_i 's. They involve the Oseen tensor, that is the fundamental solution of the Stokes operator in $\mathbb{R}^3$, and its derivatives. They are not fitted to our half-space case, as the inhomogeneous Dirichlet data that they create at the boundary is too large. We must therefore adapt this strategy, constructing the analogue of point Stokeslets for a half-space with homogeneous Dirichlet condition. We believe that this part of the analysis is of independent interest, and contributes to the interesting problem of extending the method of reflections to domains with boundaries [23].

Step 2 : wall law. — Once the continuous model (7) has been derived, the goal is to understand the effect of the depletion layer D^ε on the solution u^ε . In other words, one would like to improve the crude approximation

$$(9) \quad \begin{aligned} -\Delta u^0 + \nabla p^0 &= f - \rho e & \text{in } \Omega, \\ \operatorname{div} u^0 &= 0 & \text{in } \Omega^0, \\ u^0|_{\partial\Omega^0} &= 0, \end{aligned}$$

where the impact of the layer is totally neglected. In this regard, we will prove in Section 4 the following theorem:

THEOREM 1.2. — *Assume that f and ρ are smooth, compactly supported in $\overline{\Omega^0}$. Then, for all $m \in \mathbb{N}$, one has*

$$u^\varepsilon = u^0 + \varepsilon u^1 + \cdots + \varepsilon^m u^m + O(\varepsilon^{m+1}) \quad \text{in } \dot{H}^1(\Omega^0),$$

where u^0 solves (9), and for each $i \geq 1$, u^i solves an homogeneous Stokes equation in Ω^0 with inhomogeneous Dirichlet data at $\partial\Omega^0$ coming from lower order profiles.

This theorem provides an asymptotic expansion of u^ε at arbitrary order, where each successive correction refines the description of the effect of the depletion layer. One could relax the assumptions on the smoothness of ρ and f : for instance, it is enough that $\nabla u^0, p^0$ belong to $H^{m+2}(\Omega^0)$ to have an expansion with m terms.

The analysis leading to Theorem 1.2 shows in particular that the inhomogeneous Dirichlet data for u^1 is given by

$$(10) \quad u^1|_{\partial\Omega^0} = (0, \partial_1 u_2^0, \partial_1 u_3^0)|_{\partial\Omega^0}.$$

This will imply, see Section 4:

COROLLARY 1.3. — *Let ρ and f as in the previous theorem, and u^S the solution of the Stokes system with Navier slip boundary condition:*

$$(11) \quad \begin{aligned} -\Delta u^S + \nabla p^S &= f - \rho e && \text{in } \Omega^0, \\ \operatorname{div} u^S &= 0 && \text{in } \Omega^0, \\ u^S|_{\partial\Omega^0} &= \varepsilon(0, \partial_1 u_2^S, \partial_1 u_3^S)|_{\partial\Omega^0}. \end{aligned}$$

Then, one has the estimate

$$\|u^\varepsilon - u^S\|_{\dot{H}^1(\Omega^0)} = O(\varepsilon^2).$$

This $O(\varepsilon^2)$ estimate shows that imposing a wall law of Navier type at the artificial boundary $\partial\Omega^0$ improves the $O(\varepsilon)$ error estimate provided by the homogeneous Dirichlet condition. It is the rigorous translation of the notion of apparent slip evoked in several papers, [1, 27, 16]. We will further discuss it in Section 5. The other phenomenon mentioned above, that is *intrinsic convection* is more subtle and corresponds to a degenerate regime of what we consider here. We will also discuss it in Section 5.

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2. STOKES PROBLEM ON THE HALF SPACE: FUNDAMENTAL SOLUTION AND STOKESLETS

For the analysis carried in this section 2 and the next section 3, there is no restriction in taking $\varepsilon = 0$, that is working on $\Omega^0 = (0, +\infty) \times \mathbb{R}^2$. The results adapt straightforwardly to $\Omega^\varepsilon = (-\varepsilon, +\infty) \times \mathbb{R}^2$ by translation.

REMARK 3.2. — Since the Wasserstein distance W_1 corresponds to the dual norm on Lipschitz functions, one could also use the following estimate

$$\int_{\Omega^0} (\rho^N - \rho) g \cdot u_\psi \leq CW_1(\rho^N, \rho) \|u_\psi\|_{\text{Lip}},$$

assuming ρ to be a probability measure. However the norm $\|\cdot\|_{(W^{2,q'}(\Omega^0))^*}$ is more accurate since $W^{2,q'}(\Omega^0) \subset \text{Lip}(\Omega^0)$. $\square$

4. THE APPROXIMATE CONTINUOUS MODEL

The purpose of this section is to analyze system (7), and to prove Theorem 1.2 and Corollary 1.3. We assume here that f and ρ are smooth and compactly supported in $\bar{\Omega}^0$.

4.1. ASYMPTOTIC EXPANSION. — We will construct an approximation of the solution $(u^\varepsilon, p^\varepsilon)$ of (7), of the following form:

$$\begin{cases} (u_{\text{app}}^\varepsilon, p_{\text{app}}^\varepsilon)(x_1, x_2, x_3) \approx \sum_{i=0}^m \varepsilon^i (u^i, p^i)(x_1, x_2, x_3), & x = (x_1, x_2, x_3) \in \Omega^0, \\ (u_{\text{app}}^\varepsilon, p_{\text{app}}^\varepsilon)(x_1, x_2, x_3) \approx \sum_{i=0}^m \varepsilon^i (U^i, P^i)(x_1/\varepsilon, x_2, x_3), & x = (x_1, x_2, x_3) \in \Omega^\varepsilon \setminus \Omega^0. \end{cases}$$

Note that this approximation is made of two parts: an interior one, in Ω^0 , with a regular expansion in ε , and a boundary layer part, localized in the depletion layer $\Omega^\varepsilon \setminus \Omega^0$. The boundary layer expansion involves boundary layer profiles $U^i = U^i(s, x_2, x_3)$, with the variable $s \in (-1, 0)$ that stands for x_1/ε . It is also convenient to set:

$$(u^i, p^i) = 0, \quad (U^i, P^i) = 0 \quad \text{for } i < 0.$$

Plugging the interior expansion in the Stokes equation, we find: for all $i \geq 0$, in Ω^0

$$(25) \quad \begin{aligned} -\Delta u^i + \nabla p^i &= \delta_{0i}(f - \rho e), \\ \text{div } u^i &= 0. \end{aligned}$$

Plugging the boundary layer expansion in the Stokes equation, we find: for all $i \geq 0$, for all $(s, x_2, x_3) \in (-1, 0) \times \mathbb{R}^2$:

$$(26) \quad -\partial_s^2 U_1^i - (\partial_2^2 + \partial_3^2) U_1^{i-2} + \partial_s P^{i-1} = 0,$$

$$(27) \quad -\partial_s^2 U_2^i - (\partial_2^2 + \partial_3^2) U_2^{i-2} + \partial_2 P^{i-2} = 0,$$

$$(28) \quad -\partial_s^2 U_3^i - (\partial_2^2 + \partial_3^2) U_3^{i-2} + \partial_3 P^{i-2} = 0,$$

$$(29) \quad \partial_s U_1^i + \partial_2 U_2^{i-1} + \partial_3 U_3^{i-1} = 0$$

The Dirichlet condition $u_{\text{app}}^\varepsilon|_{\partial\Omega^\varepsilon} \approx 0$ further yields: for all $i \geq 0$,

$$(30) \quad U^i|_{s=-1} = 0.$$

Eventually, conditions

$$[u_{\text{app}}^\varepsilon]|_{\partial\Omega^0} \approx 0, \quad [\Sigma(u_{\text{app}}^\varepsilon, p_{\text{app}}^\varepsilon)n]|_{\partial\Omega^0} \approx 0,$$

which reflect continuity of the velocity field and the stress tensor at the interface $\partial\Omega^0$ give for all $i \geq 0$:

$$(31) \quad U^i|_{s=0} = u^i|_{x_1=0}$$

$$(32) \quad \partial_s U^i|_{s=0} - P^{i-1}|_{s=0}(1, 0, 0)^t = \partial_1 u^{i-1}|_{x_1=0} - p^{i-1}|_{x_1=0}(1, 0, 0)^t.$$

Computation of the first terms. — We compute U^0, u^0, p^0 . First, from (29) and (30), we get $\partial_s U_1^0 = 0$ and $U_1^0|_{s=-1} = 0$, which implies $U_1^0 = 0$. Then, from (27), (30) and (32), we find

$$\partial_s^2 U_2^0 = 0, \quad U_2^0|_{s=-1} = 0, \quad \partial_s U_2^0|_{s=0} = 0,$$

which leads to $U_2^0 = 0$. Similarly, $U_3^0 = 0$. Hence, $U^0 = 0$. Next, we consider (25) and (31) at rank $i = 0$. They imply that (u^0, p^0) satisfies (7). In particular, as f belongs to $L^{6/5}(\Omega_0) \cap H^m(\Omega^0)$, $(\nabla u^0, p^0)$ belongs to $H^m(\Omega^0)$ for all m .

Computation of next order terms. — We now assume that $i \geq 1$ and that profiles U^k, P^{k-1}, u^k, p^k are known for all $k \leq i-1$, with:

$$U^k, P^{k-1} \in H^\infty((-1, 0) \times \mathbb{R}^2), \quad \nabla u^k, p^k \in H^\infty(\Omega^0).$$

We now show how to construct U^i, P^{i-1}, u^i, p^i . Considering (29) and (30) yields

$$U_1^i(s, \cdot) = - \int_{-1}^s (\partial_2 U_2^{i-1} + \partial_3 U_3^{i-1})(t, \cdot) dt \in H^\infty((-1, 0) \times \mathbb{R}^2).$$

Next, we consider (26) and (32) which allow the calculation of $P^{i-1} \in H^\infty((-1, 0) \times \mathbb{R}^2)$:

$$P^{i-1}(s, \cdot) = \partial_s U_1^i|_{s=0} - \partial_1 u_1^{i-1}|_{x_1=0} + p^{i-1}|_{x_1=0} + \int_0^s (\partial_s^2 U_1^i + (\partial_2^2 + \partial_3^2) U_1^{i-2})(t, \cdot) dt.$$

Next, we consider (27)–(30)–(32): these relations imply

$$U_2^i(s, \cdot) = - \int_{-1}^s \int_0^t \left((\partial_2^2 + \partial_3^2) U_2^{i-2} + \partial_2 P^{i-2} \right) (t', \cdot) dt' dt + (s+1) \partial_1 u_2^{i-1}|_{x_1=0}.$$

A similar expression holds for U_3^i . Both $U_2^i, U_3^i \in H^\infty((-1, 0) \times \mathbb{R}^2)$. Eventually, we see by (25) and (31) that (u^i, p^i) solve a homogeneous Stokes equation with the inhomogeneous Dirichlet condition

$$u^i|_{x_1=0} = U^i|_{x_1=0}.$$

By standard regularity results for the Stokes equation, cf. [10, Th. IV.3.2 & IV.3.3], $\nabla u^i, p^i \in H^\infty(\Omega^0)$.

Equations for u^1 and u^2 . — Anticipating the discussion in Section 5, it is worth specifying the systems satisfied by u^1, p^1 and u^2, p^2 . From above equations, we deduce

$$U_1^1 = 0, \quad P^0 = -\partial_1 u_1^0|_{x_1=0} + p^0|_{x_1=0}, \quad (U_2^1, U_3^1) = (s+1)(\partial_1 u_2^0, \partial_1 u_3^0)|_{x_1=0}.$$

Hence,

$$(33) \quad \begin{aligned} -\Delta u^1 + \nabla p^1 &= 0, \\ \operatorname{div} u^1 &= 0, \\ u^1|_{x_1=0} &= (0, \partial_1 u_2^0, \partial_1 u_3^0)|_{x_1=0}. \end{aligned}$$

Then,

$$\begin{aligned} U_1^2(s, \cdot) &= - \int_{-1}^s (\partial_2 U_2^1 + \partial_3 U_3^1)(t, \cdot) dt \\ &= - \int_{-1}^s (t+1)(\partial_2 \partial_1 u_2^0 + \partial_3 \partial_1 u_3^0)|_{x_1=0} dt = \frac{(s+1)^2}{2} \partial_1^2 u_1^0|_{x_1=0}, \end{aligned}$$

while

$$\begin{aligned} (U_2^2, U_3^2) &= (s+1)(\partial_1 u_2^1, \partial_1 u_3^1)|_{x_1=0} + \frac{1}{2}(s-1)(s+1)(\partial_2 P^0, \partial_3 P^0) \\ &= (s+1)(\partial_1 u_2^1, \partial_1 u_3^1)|_{x_1=0} \\ &\quad + \frac{1}{2}(s-1)(s+1)(-\partial_1 \partial_2 u_1^0 + \partial_2 p^0, -\partial_1 \partial_3 u_1^0 + \partial_3 p^0)|_{x_1=0}. \end{aligned}$$

We recover the system

$$(34) \quad \begin{aligned} -\Delta u^2 + \nabla p^2 &= 0, \\ \operatorname{div} u^2 &= 0, \\ u_1^2|_{x_1=0} &= \frac{1}{2} \partial_1^2 u_1^0|_{x_1=0}, \\ u_2^2|_{x_1=0} &= \partial_1 u_2^1|_{x_1=0} + \frac{1}{2}(\partial_1 \partial_2 u_1^0 - \partial_2 p^0)|_{x_1=0}, \\ u_3^2|_{x_1=0} &= \partial_1 u_3^1|_{x_1=0} + \frac{1}{2}(\partial_1 \partial_3 u_1^0 - \partial_3 p^0)|_{x_1=0}. \end{aligned}$$

4.2. PROOF OF THEOREM 1.2. — We will show here that the expansion constructed in the previous section provides an approximate solution of the true solution u^ε of (7). Given $m \in \mathbb{N}$, we first introduce $\tilde{u}^{m+2}$ the solution in $\dot{H}^1(\Omega^\varepsilon)$ of the Stokes problem

$$\begin{cases} -\Delta \tilde{u}^{m+2} + \nabla \tilde{p}^{m+2} = 0 & \text{in } \Omega^\varepsilon, \\ \operatorname{div} \tilde{u}^{m+2} = -\partial_1 u_1^{m+2} 1_{\Omega^0} & \text{in } \Omega^\varepsilon, \\ \tilde{u}^{m+2}|_{\partial\Omega^\varepsilon} = 0. \end{cases}$$

Such $\tilde{u}^{m+2}$ exists, cf. [10, Th. IV.3.3], and satisfies

$$\|\nabla \tilde{u}^{m+2}\|_{L^2(\Omega^\varepsilon)} \lesssim \|\partial_1 u_1^{m+2} 1_{\Omega^0}\|_{L^2(\Omega^\varepsilon)} \lesssim 1.$$

We then define $(u_{\text{app}}^\varepsilon, p_{\text{app}}^\varepsilon)$ as follows:

$$\begin{aligned} (u_{\text{app}}^\varepsilon, p_{\text{app}}^\varepsilon)(x, y, z) &= \sum_{i=0}^{m+1} \varepsilon^i (u^i, p^i)(x, y, z) + \varepsilon^{m+2} (u_1^{m+2}, 0, 0, 0)(x, y, z) \\ &\quad + \varepsilon^{m+2} (\tilde{u}^{m+2}, \tilde{p}^{m+2})(x, y, z), \quad X = (x, y, z) \in \Omega^0, \\ (u_{\text{app}}^\varepsilon, p_{\text{app}}^\varepsilon)(x, y, z) &= \sum_{i=0}^{m+1} \varepsilon^i (U^i, P^i)(x/\varepsilon, y, z) + \varepsilon^{m+2} (U_1^{m+2}, 0, 0, 0)(x/\varepsilon, y, z) \\ &\quad + \varepsilon^{m+2} (\tilde{u}^{m+2}, \tilde{p}^{m+2})(x, y, z), \quad X = (x, y, z) \in \Omega^\varepsilon \setminus \Omega^0. \end{aligned}$$

We then introduce $v^\varepsilon = u^\varepsilon - u_{\text{app}}^\varepsilon$, $q^\varepsilon = p^\varepsilon - p_{\text{app}}^\varepsilon$, solving

$$\begin{aligned} -\Delta v^\varepsilon + \nabla q^\varepsilon &= -\varepsilon^{m+2} (\Delta u_1^{m+2}, 0, 0) \quad \text{in } \Omega^0, \\ -\Delta v^\varepsilon + \nabla q^\varepsilon &= -\sum_{i=m}^{m+1} \varepsilon^i (\partial_2^2 + \partial_3^2)(\varepsilon U_1^{i+1}, U_2^i, U_3^i) - \sum_{i=m}^{m+1} \varepsilon^i (0, \partial_2 P^i, \partial_3 P^i) \quad \text{in } D^\varepsilon, \\ (35) \quad \operatorname{div} v^\varepsilon &= 0 \quad \text{in } \Omega^\varepsilon, \\ [v^\varepsilon]|_{\partial\Omega^0} &= 0, \\ [\Sigma(v^\varepsilon, q^\varepsilon)n]|_{\partial\Omega^0} &= -\varepsilon^{m+1} (\varepsilon \partial_1 u_1^{m+2}|_{x_1=0}, \partial_1 u_2^{m+1}|_{x_1=0}, \partial_1 u_3^{m+1}|_{x_1=0}), \\ v^\varepsilon|_{\partial\Omega^\varepsilon} &= 0. \end{aligned}$$

Performing a simple energy estimate, we find

$$\int_{\Omega^\varepsilon} |\nabla v^\varepsilon|^2 = \varepsilon^{m+2} \int_{\Omega^0} \nabla u_1^{m+2} : \nabla u_1^\varepsilon - \int_{D^\varepsilon} R^\varepsilon \cdot v^\varepsilon + \int_{\partial\Omega^0} r^\varepsilon \cdot v^\varepsilon,$$

where

$$\begin{aligned} R^\varepsilon &= -\sum_{i=m}^{m+1} \varepsilon^i (\partial_2^2 + \partial_3^2)(\varepsilon U_1^{i+1}, U_2^i, U_3^i) - \sum_{i=m}^{m+1} \varepsilon^i (0, \partial_2 P^i, \partial_3 P^i) = O(\varepsilon^m), \\ r^\varepsilon &= \varepsilon^{m+1} (\varepsilon \partial_1 u_1^{m+2}|_{x_1=0}, \partial_1 u_2^{m+1}|_{x_1=0}, \partial_1 u_3^{m+1}|_{x_1=0}) = O(\varepsilon^{m+1}). \end{aligned}$$

By the Poincaré and trace inequalities

$$\|v^\varepsilon\|_{L^2(D^\varepsilon)} \lesssim \varepsilon \|\nabla v^\varepsilon\|_{L^2(D^\varepsilon)}, \quad \|v^\varepsilon\|_{L^2(\partial\Omega^0)} \lesssim \sqrt{\varepsilon} \|\nabla v^\varepsilon\|_{L^2(D^\varepsilon)},$$

we infer easily

$$\|\nabla v^\varepsilon\|_{L^2(\Omega^\varepsilon)} \lesssim \varepsilon^{m+1}.$$

Restricting to Ω^0 , Theorem 1.2 follows.

4.3. WALL LAW OF NAVIER TYPE : PROOF OF COROLLARY 1.3. — We conclude Section 4 by proving Corollary 1.3. We know from (1.2) that

$$u^\varepsilon = u^0 + \varepsilon u^1 + O(\varepsilon^2) \quad \text{in } \dot{H}^1(\Omega^0).$$

Hence, Corollary 1.3 will follow from the estimate

$$\|\nabla(u^S - u_{\text{app}}^S)\|_{L^2(\Omega^0)} = O(\varepsilon^2), \quad u_{\text{app}}^S = u^0 + \varepsilon u^1.$$

together with the following estimates

$$|A(x, y)| \leq C \left(\frac{|x_1|}{|x - y^I|} + 1 \right) \left(\frac{1}{|x - y^I|} + \frac{|y_1|}{|x - y^I|^2} \right),$$

$$|\nabla_x A(x, y)| + |\nabla_y A(x, y)| \leq C \left(\frac{|x_1|}{|x - y^I|} + 1 \right) \left(\frac{1}{|x - y^I|^2} + \frac{|y_1|}{|x - y^I|^3} \right). \quad \square$$

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